The Relationship Between Cocoa Cultivation and Deforestation in Cameroon (2001–2023)

IB Environmental Systems and Societies — Standard Level Individual Investigation

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Table of Contents

§	Section	Page
1	Introduction and environmental context	[p]
2	Research question and rationale	[p]
3	Strategy and perspectives	[p]
4	Methodology	[p]
5	Data presentation and processing	[p]
6	Analysis and discussion	[p]
7	Conclusion	[p]
8	Evaluation	[p]
9	References	[p]
10	Appendices	[p]

Page numbers to be completed after final pagination in the submitted document.

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1. Introduction and environmental context

Deforestation is one of the clearest cases of human activity altering an environmental system faster than that system can recover. Removing closed-canopy tropical forest reduces at once the carbon held in biomass, the habitat supporting high species richness, the regulation of local rainfall and the binding of soil against erosion. Forest is a store of natural capital and the services it provides are natural income. Clearing it replaces a largely self-regulating system, held near a steady state by internal negative feedback, with a simplified agricultural system that stays productive only through continuous human input. Recovery is slow: regrowth may restore canopy within decades but not the species composition or soil structure of primary forest.

Cameroon lies at the north-western edge of the Congo Basin, the world's second largest block of tropical rainforest. Its southern and south-western regions hold dense humid forest and grow almost all of the country's cocoa, so agricultural and forest systems overlap spatially. Tyukavina et al. (2018) found that forest disturbance across the Congo Basin between 2000 and 2014 was dominated by small-scale, largely manual clearing for agriculture rather than by industrial logging, and that the annual rate of this clearing almost doubled over the period. That is directly relevant, because Cameroonian cocoa is produced across a very large number of smallholdings rather than on estates.

Cocoa matters environmentally for two reasons. First, it is a perennial tree crop, so planting is a long-term land-use commitment and the land is unlikely to return to forest. Ordway et al. (2017) identified cocoa as the fastest-expanding export crop in sub-Saharan Africa, with most expansion occurring outside industrial plantations. Second, cocoa in Cameroon is widely grown beneath retained or planted shade trees, so an established farm can still carry a partial canopy and support moderate biodiversity. Forest converted to shaded cocoa is therefore degraded rather than cleared outright. This makes the link between cocoa and measured forest loss genuinely uncertain, and worth testing rather than assuming.

2. Research question and rationale

To what extent is the expansion of cocoa cultivation associated with annual tree-cover loss in Cameroon between 2001 and 2023?

The question is worth asking because the assumption that cocoa drives deforestation is widespread and now underpins trade policy, yet is rarely tested against national data. If the association is weak, policy aimed only at cocoa will not protect Cameroon's forests.

Independent variable: the annual change in cocoa harvested area in Cameroon (Δ hectares per year), derived from FAOSTAT. Harvested area is used rather than production because tonnage varies with rainfall, disease and input use rather than with land occupied.

Dependent variable: annual tree-cover loss in Cameroon (hectares per year) from the UMD/Global Forest Watch dataset, at a fixed 30% tree-canopy-density threshold.

Controlled aspects: a single country boundary, one canopy threshold, one dataset version, and the same calendar years for both variables throughout.

One refinement is necessary. Harvested area is a *stock* — the area standing in a given year — whereas tree-cover loss is a *flow*. Correlating a stock against a flow when both rise over 23 years returns a coefficient generated mainly by shared time trend. Defining "expansion" as the **annual change** in harvested area makes both variables flows in ha yr^{-1} over matching years, the defensible comparison. The question itself is unchanged.

Boundaries. Cameroon is used as the spatial unit because both datasets report to national boundaries, allowing consistent matching. The period 2001–2023 is set by the tree-cover-loss record, which begins in 2001; FAOSTAT extends further back but is truncated to match.

3. Strategy and perspectives

This investigation sits inside a genuine conflict rather than a technical puzzle, and the groups involved hold different environmental value systems which lead them to read the same forest differently.

Smallholder farmers. For the households that grow Cameroon's cocoa, forest is a land reserve and cocoa is one of the few reliable sources of cash income in rural areas. Clearing a hectare is rational for a household with limited capital: forest soils are fertile, the land is often effectively free to occupy, and a cocoa farm yields income for decades. This anthropocentric position is not an uninformed one — it reflects real constraints rather than indifference to forests.

The national economy. Cocoa is a significant source of export earnings and employment, so the state has an interest in sustained output. This creates a tension within government, since the same state holds forest-protection commitments. Restricting cultivation carries a measurable cost in foreign-exchange earnings, which is why forest policy is often weakly enforced.

Conservation interests. From an ecocentric standpoint, Congo Basin forest has value as habitat, as a carbon store and as a regulator of regional hydrology — value largely non-monetised and accruing globally rather than locally. That asymmetry is central: the benefits of clearing are captured locally and immediately, while the costs of lost ecosystem services are global and deferred. This externality explains why the issue is not resolved by farmers simply choosing better.

Three strategies are advanced, each with real drawbacks.

Agroforestry. Growing cocoa under shade retains part of the canopy, supports more biodiversity than full-sun monoculture, stores more carbon and improves soil protection. It is the dominant system in Cameroon already, which is an advantage. Its weakness is yield: shaded cocoa produces less per hectare, so if demand is fixed, lower yields require *more* land. This is the land-sharing versus land-sparing tension. Gockowski and Sonwa (2011) modelled the alternative and estimated that if intensified cocoa technology had been adopted from the 1960s across Côte d'Ivoire, Ghana, Nigeria and Cameroon, more than 21,000 km^2 of deforestation and forest degradation and roughly 1.4 billion tonnes of CO_2 emissions could have been avoided. Their conclusion is uncomfortable for conservation thinking: intensification, though it reduces on-farm biodiversity, may spare more forest overall. Which strategy is "better" depends on whether value is placed on biodiversity within the farm or on the total forest left standing.

Certification. Schemes such as Rainforest Alliance and Fairtrade aim to reward deforestation-free production with price premiums. They give consumers leverage and can transfer income, but depend on audit quality, cover only part of production, and are vulnerable to leakage: if certified farms stop expanding while demand continues, clearing moves to uncertified farms. Premiums also often fail to reach the farmer in full.

Regulation. The EU Deforestation Regulation (European Union, 2023) requires importers to show that commodities including cocoa were not grown on land deforested after 2020, with geolocation data. This is the strongest instrument available, but places the compliance burden on smallholders least able to carry it. The likely outcome is less deforestation but also exclusion

of the poorest farmers from the EU market, raising a question of equity: who pays to protect a global public good?

These perspectives shape outcomes directly: because the most economically rational household choice removes forest, and the resulting costs fall largely outside Cameroon, forest loss continues even where all parties agree it is undesirable.

4. Methodology

This investigation uses **exclusively secondary quantitative data**: no fieldwork, interviews, questionnaires or experiments were involved. The method is a reproducible desk-based correlational analysis of two independent published datasets.

Data sources.

Variable	Source	Domain / dataset	Unit	Coverage used
Cocoa harvested area	FAO (FAOSTAT)	Crops and livestock products (QCL); Cameroon (area 32), Cocoa beans (item 661), element 5312	hectares (ha)	2001–2023
Cocoa production (context only)	FAO (FAOSTAT)	as above, element 5510	tonnes (t)	2001–2023
Annual tree-cover loss	UMD / Global Forest Watch	Global Forest Change, annual tree-cover loss, 30% canopy threshold	hectares (ha)	2001–2023

Extraction procedure. From the FAOSTAT QCL interface, Cameroon was selected as the country, *Cocoa beans* as the item, and *Area harvested* and *Production* as elements, for 2001–2023, with flags and units displayed, and exported as CSV. From the Global Forest Watch Cameroon dashboard, the annual tree-cover-loss widget was set to a **30% tree-canopy-density threshold** and exported as CSV. Both files were imported unaltered and all processing done in a scripted pipeline, so every published figure traces back to a raw file. Dataset version and access date were recorded for both sources.

Processing. Harvested area was converted into its annual change, $\Delta A_t = A_t - A_{t-1}$, giving the independent variable in ha yr^{-1} for 2002–2023 (n = 22; one year is necessarily lost). Percentage

change was calculated for both series to allow comparison of relative rather than absolute movement.

Statistical treatment. Spearman's rank correlation (r_s) is the primary test. It is preferred to Pearson's because it needs only a monotonic relationship, assumes no bivariate normality, and resists the outliers expected in a series containing estimated and imputed values. Pearson's *r* is also reported, with a Shapiro–Wilk test of normality to show whether its assumptions hold. For each coefficient the sample size, two-tailed *p*-value and the critical value at $\alpha = 0.05$ are reported, because with $n \approx 22$ statistical power is low and a coefficient must exceed approximately 0.43 to reach significance.

Three further tests address specific threats to validity rather than being added for their own sake:

1. **First-differenced correlation** — both series differenced to remove shared time trend, testing whether any association survives detrending or is simply two rising lines.
2. **Lagged correlation (lags 0–5 years)** — because FAO defines harvested area as the area actually gathered, which for a tree crop excludes immature plantings. Cocoa bears fruit 3–5 years after planting, so clearing in year *t* should appear in harvested area around $t + 4$.
3. **Sub-period analysis (2011–2023)** — because Global Forest Watch states that the 2011-onward loss data use an updated and more sensitive algorithm, so 2001–2023 is not fully internally consistent.

Reliability and validity assessment. Each FAOSTAT year was checked for its flag and the proportion of non-official values recorded. The two sources are methodologically independent — survey reporting versus satellite classification — so neither propagates the other's error, but equally they were never designed to be compared.

5. Data presentation and processing

> **STATUS — awaiting data.** The two source files were not retrievable from this environment (network policy blocked `fao.org` and `globalforestwatch.org`; see the accompanying data verification report). Tables 1–3 and Figures 1–3 are produced automatically by `analysis/ess_analysis.py` once the CSVs are placed in `data/`. **No values have been invented to fill them.**

Table 1. Cocoa harvested area, cocoa production and annual tree-cover loss, Cameroon, 2001–2023.

Year	Cocoa harvested area (ha)	FAO flag	Annual change Δ (ha)	Cocoa production (t)	Tree-cover loss (ha)
2001			—		
…					
2023					

Sources: FAO (2026), FAOSTAT QCL, element 5312/5510, item 661, area 32; Global Forest Watch (2026), UMD Global Forest Change, 30% canopy threshold. Units as stated in column headings.

Table 2. Percentage change, period summary statistics (mean, median, standard deviation, range) for both variables, and sub-period means for 2001–2010 and 2011–2023.

Table 3. Correlation results: Spearman's r_s and Pearson's r with n, two-tailed p and critical values, for (a) levels, (b) first differences, (c) lags 0–5, (d) 2011–2023 only.

Figure 1. Cocoa harvested area (ha) and annual tree-cover loss (ha), Cameroon, 2001–2023 — dual-axis time series. **Figure 2.** Scatter plot of annual tree-cover loss against annual change in cocoa harvested area, with least-squares line and r/r_s annotated. **Figure 3.** Lagged correlation coefficient against lag (0–5 years), with the $\alpha = 0.05$ critical value marked.

All processing steps are scripted and the script is reproduced in Appendix C, so every figure in this report can be traced to a raw downloaded file. Calculations are shown in Appendix B.

6. Analysis and discussion

> **STATUS — numerical results pending.** Placeholders of the form `{{...}}` are filled automatically by the analysis script from the downloaded files. The analytical reasoning, the confounders and the interpretation framework below are complete and do not depend on the values.

Trends in the two variables. Over 2001–2023 cocoa harvested area moved from `{{AREA_2001}}` ha to `{{AREA_2023}}` ha, a change of `{{AREA_PCT}}`%, while annual tree-cover loss moved from `{{LOSS_2001}}` ha to `{{LOSS_2023}}` ha (`{{LOSS_PCT}}`%). Mean annual loss was `{{LOSS_MEAN_0110}}` ha for 2001–2010 against `{{LOSS_MEAN_1123}}` ha for 2011–2023. That step must be read cautiously: Global Forest Watch attributes part of it to the more sensitive algorithm introduced for 2011 rather than to change on the ground.

Correlation results. Against the annual change in harvested area, tree-cover loss gave r_s = `{{RHO_DIFF}}` (p = `{{P_DIFF}}`, n = `{{N_DIFF}}`) and r = `{{R_DIFF}}`. The levels-versus-levels comparison gave r_s = `{{RHO_LEVEL}}` (p = `{{P_LEVEL}}`), but this is the trend-confounded pairing identified in §2, reported only for completeness: because both series rise, a positive coefficient is expected even if the variables are unrelated. The first-differenced result carries the evidential weight, since differencing removes the shared trend and tests whether years of faster cocoa expansion were also years of greater loss.

Lag structure. The strongest coefficient occurred at a lag of `{{BEST_LAG}}` years (r_s = `{{BEST_LAG_RHO}}`, p = `{{BEST_LAG_P}}`). A peak at three to five years would be consistent with the biology of the crop, since land cleared in one year only enters FAO's harvested-area figures once the trees begin bearing. A peak at zero or at an implausibly long lag would instead suggest noise. Six lags were tested on roughly twenty observations, which inflates the chance of a spurious "best" result, so this is treated as exploratory rather than confirmatory.

Does the evidence support an association? `{{VERDICT_SENTENCE}}` A coefficient must exceed about 0.43 for significance here, so the design detects only a fairly strong relationship; a weak but real association would be indistinguishable from chance. A non-significant result is

therefore evidence of a weak *measurable* link at national scale, not proof that cocoa has no effect on forests.

Why the association is likely to be diluted. Several mechanisms work against detecting a clear signal. Tree-cover loss aggregates every cause of canopy removal. Curtis et al. (2018) attribute only about 27% of global forest loss to permanent conversion for commodity production, with forestry, shifting agriculture and fire accounting for most of the rest. In Cameroon, loss includes selective logging in the eastern concessions, clearing for food crops such as plantain, maize and cassava, fuelwood and charcoal extraction, oil palm expansion, road and mining development, and fire in the drier north — none of it cocoa. Tyukavina et al. (2018) found Congo Basin disturbance dominated by small-scale clearing for agriculture generally, not by one export crop. Cocoa is one contributor competing with several larger ones in the same national total.

The agroforestry point sharpens this. Because much Cameroonian cocoa is shaded, conversion may leave canopy density above the 30% threshold and so be partly invisible to the algorithm. The measured relationship would then understate real land-use change — a measurement system valid for canopy monitoring but only partly valid for the question asked here.

Alternative explanations. Any positive association could arise without cocoa causing forest loss. Both variables respond to rural population growth and road access, which simultaneously raise clearing of all kinds and extend cultivated area. World cocoa prices affect planting decisions and could drive expansion in years that happen to coincide with loss from other causes. Conversely, causation could run the other way at farm level: households already clearing forest for food crops may plant cocoa afterwards, making cocoa a consequence of clearing rather than its cause. The design used here cannot separate these, so **no causal claim is made.** Correlation between two national annual series, even where significant, establishes temporal co-movement only.

7. Conclusion

> **STATUS — to be finalised from the computed results.**

Answering the research question directly: the expansion of cocoa cultivation is associated with annual tree-cover loss in Cameroon between 2001 and 2023 **to a `{{STRENGTH_WORD}}` extent**. The primary test, relating annual change in cocoa harvested area to annual tree-cover loss, returned r_s = `{{RHO_DIFF}}` (p = `{{P_DIFF}}`, n = `{{N_DIFF}}`), `{{SIG_PHRASE}}` at α = 0.05. The trend-confounded levels comparison returned a `{{LEVEL_DESCRIPTOR}}` coefficient, which illustrates how much apparent association in this topic comes from two variables rising together rather than from any direct link.

The objectives were partly achieved. The investigation succeeded in assembling two independent, documented national datasets on a common annual basis and testing them with appropriate non-parametric statistics, and it succeeded in showing that a simple "cocoa causes deforestation" reading is not supported by national aggregate data. It did not succeed in isolating cocoa's contribution to forest loss, which national-scale correlation cannot do. The defensible conclusion is that cocoa cultivation is **one** pressure on Cameroonian forest among several, and that at national resolution its signal is `{{SIGNAL_DESCRIPTOR}}` relative to the

combined effect of logging, food-crop clearing and other drivers. Every statement here rests on the data presented; nothing is extrapolated beyond them.

8. Evaluation

Reliability of the sources. The two datasets are reliable for their own purposes but unequally so for this one. Global Forest Watch rests on a consistent, peer-reviewed satellite method (Hansen et al., 2013) and is reproducible: the same imagery and algorithm yield the same result. FAOSTAT depends on national reporting, and `{{FLAG_SUMMARY}}` of the Cameroon cocoa harvested-area values carry FAO flags indicating estimated or imputed rather than officially reported figures. Imputed values are typically carried forward or interpolated, which **artificially smooths the series** and suppresses exactly the year-to-year variation the correlation depends on — the most serious limitation on the independent variable.

Validity and consistency of measurement. The central validity problem is that tree-cover loss is not deforestation: it records canopy removal regardless of cause or permanence, including plantation harvest, fire, storm damage and shifting cultivation followed by regrowth. Treating it as deforestation would overstate permanent conversion. The terms are kept separate throughout, but the dependent variable remains a proxy rather than a direct measure. A second inconsistency is the 2011 break: the post-2011 algorithm detects more loss, so part of the measured increase is an artefact, which is why 2011–2023 was analysed separately. A third is the canopy threshold; 30% is conventional but arbitrary, and another value would give different absolute losses.

Limitations of national-level data. This is the most important structural weakness. Cocoa is concentrated in the Centre, South, South-West and Littoral regions, whereas loss occurs nationwide, including logging in the East and fire in the drier north. Aggregating to the national total averages cocoa landscapes together with landscapes where cocoa is absent, diluting any genuine local relationship, and inferring farm-level behaviour from national aggregates risks the ecological fallacy.

Sample size and the limits of correlation. Twenty-two to twenty-three annual observations give low statistical power, so only a strong association is detectable, and testing six lags on so few points raises the chance that the best-fitting lag is coincidental. Correlation cannot establish causal direction or exclude a third variable driving both, and is weakened further because the two series come from incompatible methods never designed for comparison.

Confounding variables not controlled include rural population growth, road construction, world cocoa prices, timber concession activity, oil palm expansion, fuelwood demand, rainfall variability and conflict-driven displacement in the North-West and South-West, all of which can affect both variables independently.

Improvements. The strongest single improvement would be to work at sub-national scale, correlating loss within the main cocoa regions against regional cocoa statistics, removing most of the dilution described above. Second, using FAO's *area planted* rather than *area harvested*, where available, would eliminate the lag artefact created by immature trees. Third, restricting the dependent variable to Global Forest Watch's **primary humid forest loss** layer rather than all tree-cover loss would exclude plantation harvest and regrowth cycles and bring the measure

closer to true deforestation. Fourth, using only 2011–2023 throughout would remove the algorithm break at the cost of sample size.

Further research. Attribution is better answered by spatial overlay than by correlation: combining a cocoa-distribution map with annual loss pixels would show directly how much loss fell within cocoa-expansion areas. FAO-supported work mapping Cameroon's cocoa areas from satellite imagery is directly relevant, since a reliable cocoa map is the missing input for this analysis. Comparing Cameroon with Côte d'Ivoire and Ghana, where cocoa is predominantly full-sun and forest loss is higher, would also test whether shaded cultivation genuinely reduces forest loss per tonne produced.

9. References

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10. Appendices

Appendix A — Complete raw secondary datasets. Unaltered FAOSTAT QCL extract (Cameroon, cocoa beans, area harvested and production, 2001–2023, with flags) and unaltered Global Forest Watch annual tree-cover-loss extract (Cameroon, 30% canopy threshold), each with source URL, dataset version and date of access.

Appendix B — Statistical calculations. Worked example of the Spearman rank calculation for the primary test (ranking, rank differences, Σd^2, substitution into $r_s = 1 - 6\Sigma d^2/[n(n^2-1)]$); full correlation matrix; Shapiro–Wilk normality output; critical-value table for n = 22 and n = 23 at $\alpha = 0.05$ and $\alpha = 0.01$.

Appendix C — Data extraction and processing record. Exact FAOSTAT query parameters (area 32, item 661, elements 5312 and 5510), Global Forest Watch widget settings and canopy threshold, dataset version strings, dates of access, and the full analysis script used, so the entire procedure can be repeated.

Appendix D — Supplementary graphs. Cocoa yield (kg ha^{-1}) over the period; percentage-change series for both variables; 2001–2010 and 2011–2023 scatter plots shown separately to display the effect of the methodological break.

Word count note

The countable word count excludes, in line with IB requirements: the title page, table of contents, section headings, all tables and their captions, all figures and their captions, equations and calculations, in-text citations, the reference list and the appendices. `scripts/wordcount.py` reproduces the figure.